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Andrews et al.

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(54) **PROCESS FOR PREPARING TAPINAROF**

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(2013.01); *C07C 69/716* (2013.01); *A61K*
31/05 (2013.01)

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(58) **Field of Classification Search**

CPC *C07C 39/08*; *C07C 37/002*; *C07C 37/07*;
C07C 45/54; *C07C 45/63*; *C07C 45/72*;
C07C 67/00

See application file for complete search history.

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patent is extended or adjusted under 35
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(63) Continuation of application No. 16/849,346, filed on
Apr. 15, 2020, now Pat. No. 10,961,175, which is a
continuation of application No. 16/189,268, filed on
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(52) **U.S. Cl.**

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(2013.01); *C07C 50/24* (2013.01); *C07C 51/09*
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(57) **ABSTRACT**

The present invention provides processes for the preparation
of 3, 5-Dihydroxy-4-isopropyl-trans-stilbene or a salt or
solvate thereof and novel intermediates used therein. In
some embodiments the 3, 5-Dihydroxy-4-isopropyl-trans-
stilbene is prepared from (E)-2-chloro-2-isopropyl-5-styryl-
cyclohexane-1,3-dione. Also disclosed are crystal forms of
3, 5-Dihydroxy-4-isopropyl-trans-stilbene or a salt or sol-
vate thereof and pharmaceutical compositions comprising
same.

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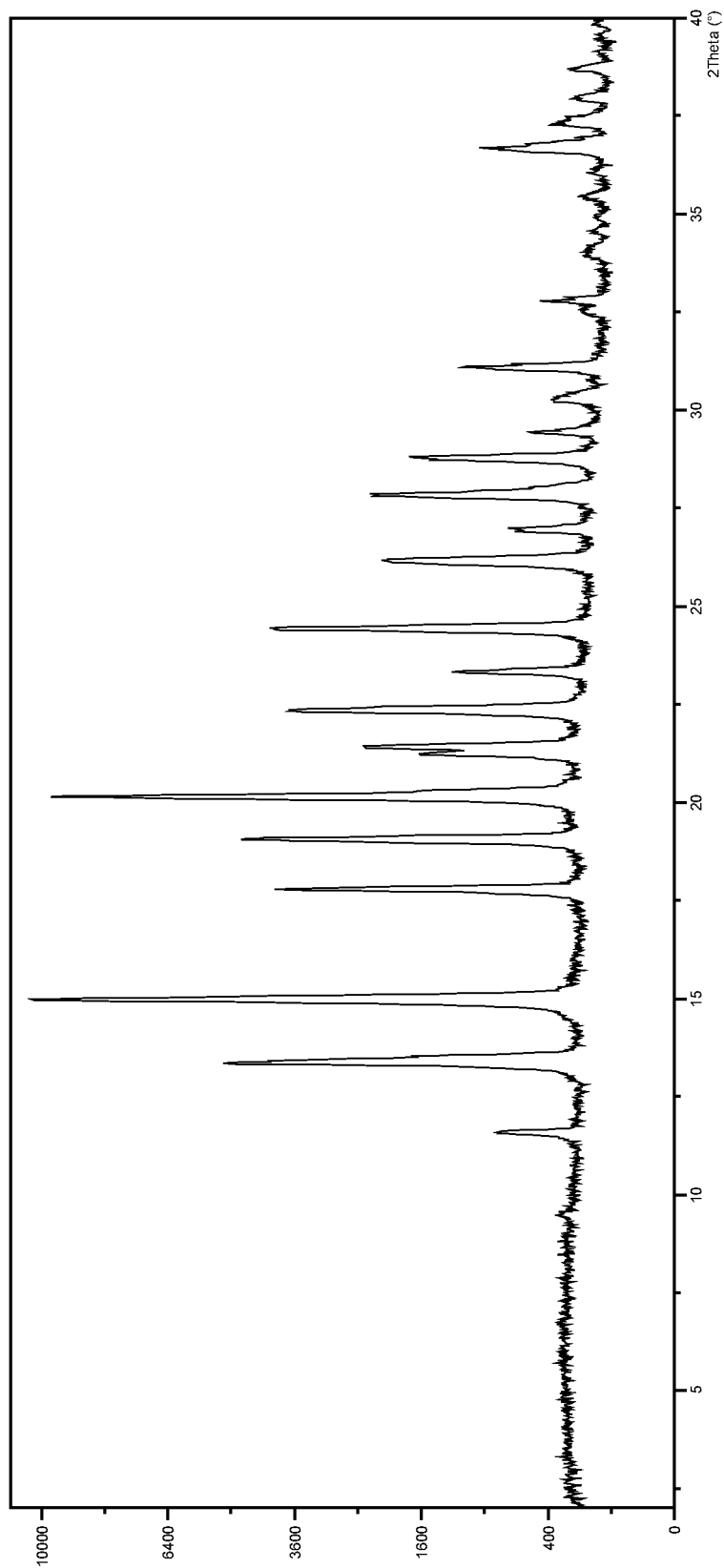


Figure 1

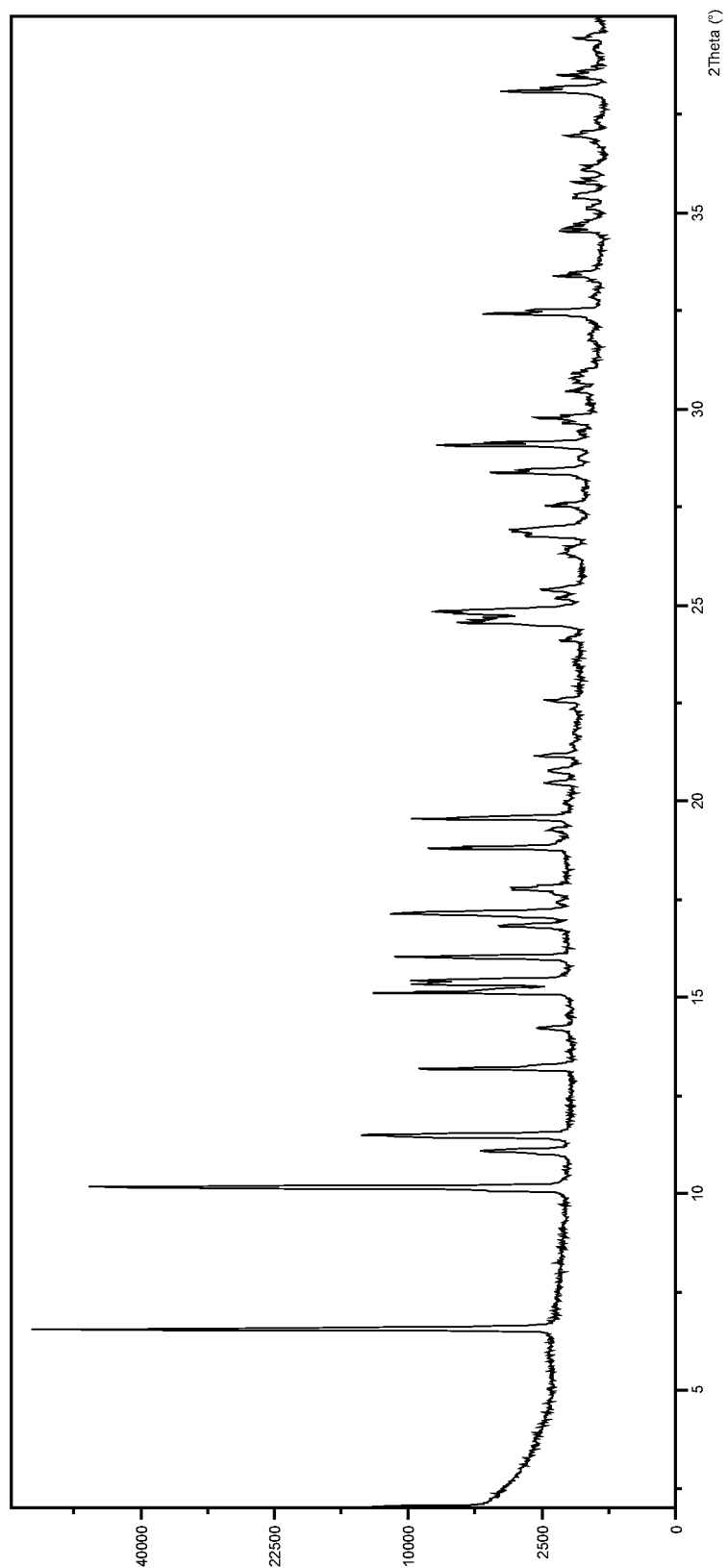


Figure 2

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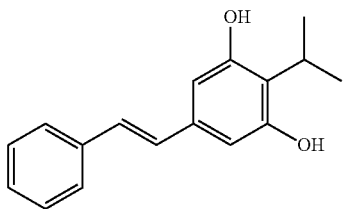
PROCESS FOR PREPARING TAPINAROF

CROSS REFERENCE TO RELATED APPLICATION

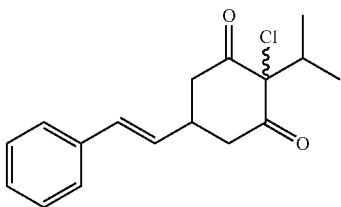
The present application is a Continuation of U.S. application Ser. No. 16/849,346, filed Apr. 15, 2020, now U.S. Pat. No. 10,961,175, which is a Continuation of U.S. application Ser. No. 16/189,268, filed Nov. 13, 2018, now U.S. Pat. No. 10,647,649, which claims benefit of and priority to U.S. Provisional No. 62/584,192 entitled "PROCESS," filed Nov. 10, 2017, the contents of which are herein incorporated by reference in their entirety.

SUMMARY OF THE INVENTION

Some embodiments of the present invention provides a processes for the preparation of the compound of Formula (I) or a salt thereof and to novel intermediates used therein

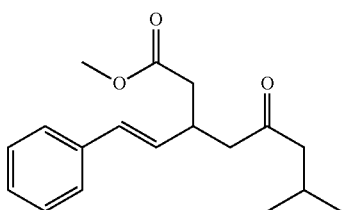


Some embodiments of the present invention describe a compound of Formula (IIa) or a salt thereof



and processes for preparing same.

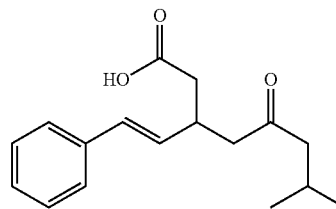
Some embodiments of the present invention describe a compound of Formula (IVa) or a salt thereof



and processes for preparing same.

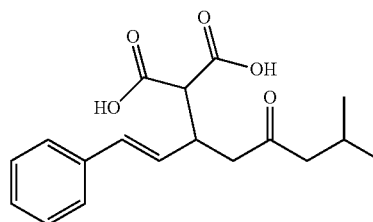
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Some embodiments of the present invention describe a compound of Formula (V) or a salt thereof



and processes for preparing same.

Some embodiments of the present invention describe a compound of Formula (VI) or a salt thereof



and processes for preparing same.

Some embodiments of the present invention describe a pharmaceutical composition which comprises a compound of Formula (I) or a salt or solvate thereof prepared according to the processes of the present invention and a pharmaceutically acceptable excipient.

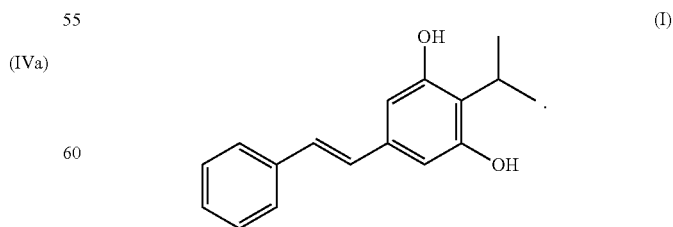
BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 shows an X-ray powder diffraction (XRPD) pattern of the compound of Formula (I) in crystalline solid state form (Form 1).

FIG. 2 shows an X-ray powder diffraction (XRPD) pattern of an acetic acid solvate of the compound of Formula (I) in crystalline solid state form.

DETAILED DESCRIPTION OF THE INVENTION

3, 5-Dihydroxy-4-isopropyl-trans-stilbene, also known as (E)-2-isopropyl-5-styrylbenzene-1,3-diol or tapinarof, of Formula (I) is a natural product derived from bacteria



This compound has several potentially useful biological responses including antibacterial, antioxidant and anti-cancer activity. PCT patent application WO200142231 dis-

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closes polyhydroxystilbenes and stilbene oxides as antipso-
riatic agents and protein kinase inhibitors.

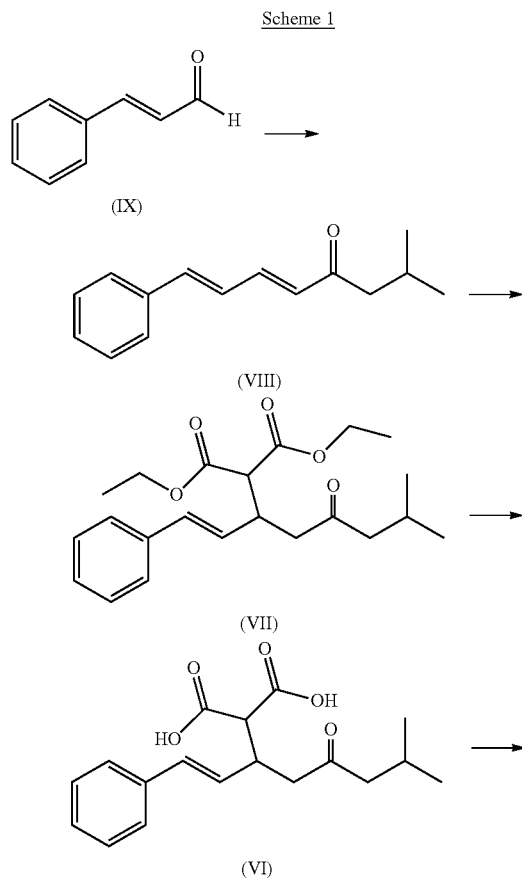
Several routes to the synthesis of the compound of
Formula (I) are known in the art. Chinese patent application
CN101648851 describes a route of synthesis via the formation
of the (E)-olefin via Horner-Wadsworth-Emmons ole-
fination. Routes are also known which are via the introduc-
tion of the isopropyl group onto a substituted resorcinol
derivative via Friedel-Crafts alkylation.

An alternative synthetic approach, similar to known bio-
synthesis route, was described by Kronenwerth et al. (*Eur. J.*
Org. Chem. 2014, 8026-8028). In this route, however, there
are problems throughout the synthesis with low yields of the
intermediates and the final product. There is also a toxicity
problem associated with incomplete removal of trace
amounts of mercury in isopropylstilbene which makes this
route unsuitable for large scale manufacturing of a pharma-
ceutical product.

Schamp et al. (*Tetrahedron*, 1973, 29, 3857-3859) dis-
closes a synthesis of simple 2-substituted resorcinols such as
2-methyl-, 2-benzyl- and 2-acetylresorcinol from the cor-
responding 1,3-cyclohexanediones. The reported conditions
involve the chlorination of the 1,3-cyclohexanediones fol-
lowed by elimination of HCl upon heating with a 25%
solution of dry hydrogen chloride in dimethylformamide.

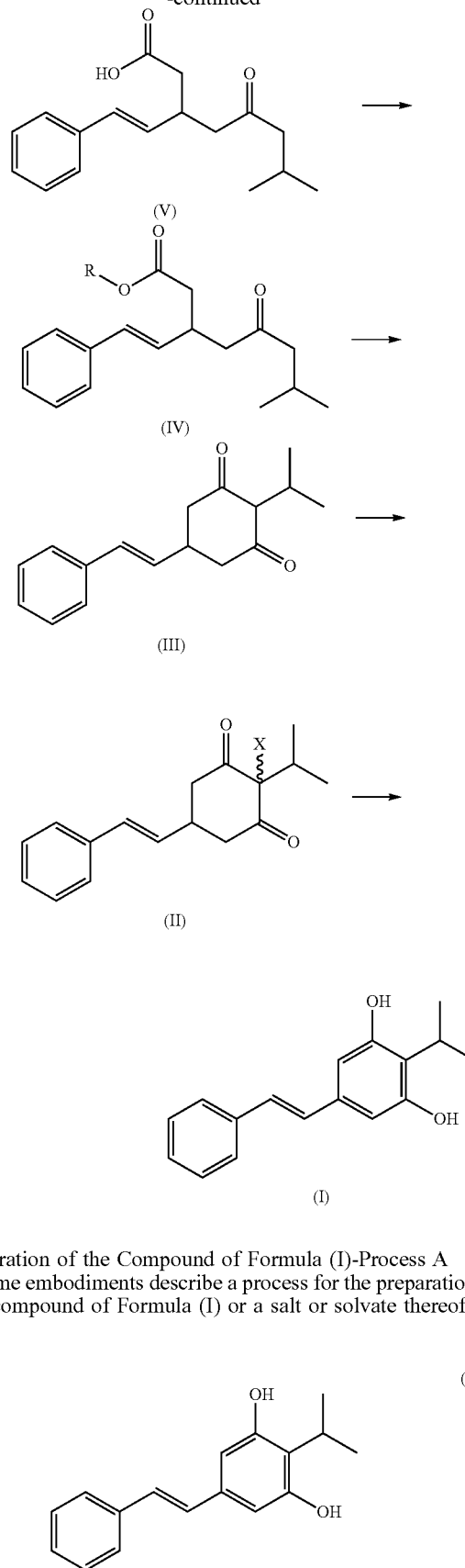
There exists a need for an efficient and scalable route for
the large scale manufacturing of the compound of Formula
(I), in particular a high yielding synthesis with no potential
toxicity problems.

The present invention provides a number of embodiments
relating to a process which is summarised in Scheme 1
below:



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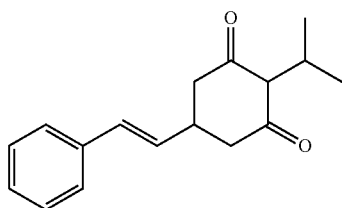
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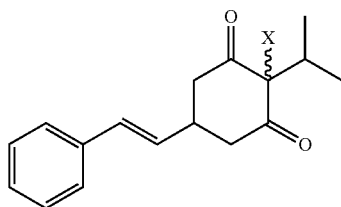
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comprising one or more of the process steps (a), (b) and (c) wherein:

(a) comprises reaction of a compound of Formula (III) or a salt thereof

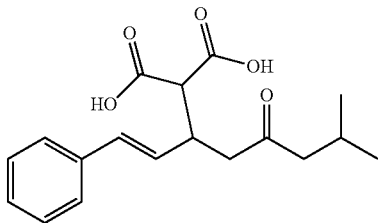


to form a compound of Formula (II) or a salt thereof

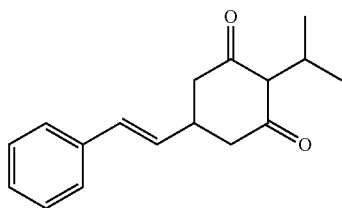


wherein X is Cl, Br, or I, and thereafter (ii) conversion of the compound of Formula (II) or a salt thereof into the compound of Formula (I) or a salt or solvate thereof;

(b) comprises conversion of a compound of Formula (VI) or a salt thereof



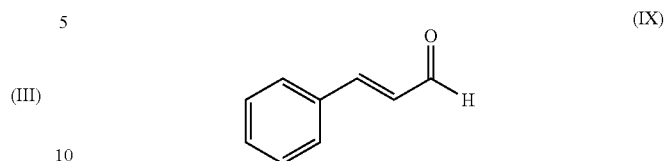
into a compound of Formula (III) or a salt thereof



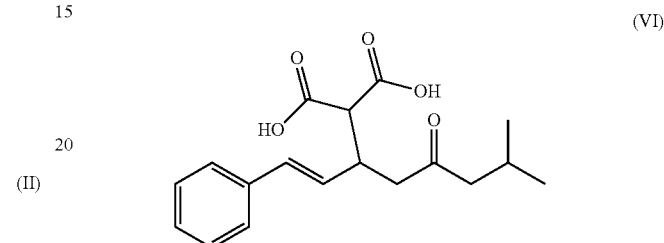
and thereafter conversion of the compound of Formula (III) or a salt thereof into a compound of Formula (I) or a salt or solvate thereof, and

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(c) comprises conversion of a compound of Formula (IX) or a salt thereof



into a compound of Formula (VI) or a salt thereof



and thereafter conversion of the compound of Formula (VI) or a salt thereof into a compound of Formula (I) or a salt or solvate thereof.

Some embodiments describe processes for making compounds of Formula (I) or a salt or solvate thereof comprising at least one of process steps (a) to (c). The processes of the present invention may include one, two or all three of process steps (a), (b) and (c).

The product of each of process steps (a), (b) and (c) may optionally be crystallised.

In some embodiments the compound of Formula (I) prepared by process (a), (b) or (c) is in crystalline solid state form. In one embodiment, there is provided, the compound of Formula (I) in crystalline solid state form (Form 1) which has an X-ray powder diffraction pattern substantially as shown in FIG. 1. In another embodiment there is provided the compound of Formula (I) in crystalline solid state characterised by an X-ray powder diffraction (XRPD) pattern with specific peaks at 15.0, 17.8, 19.1, 20.2, 21.5, 22.4, 23.3, 24.5, 26.2 and 27.9 degrees (all values $\pm 0.1^\circ$ 2θ experimental error). In another embodiment there is provided the compound of Formula (I) in crystalline solid state characterised by an X-ray powder diffraction (XRPD) pattern with at least nine, or at least eight, or at least seven, or at least six, or at least five, or at least four, or at least three specific peaks selected from 15.0, 17.8, 19.1, 20.2, 21.5, 22.4, 23.3, 24.5, 26.2 and 27.9 degrees (all values $\pm 0.1^\circ$ 2θ experimental error).

Process Step (a)

In some embodiments X is Cl.

In some embodiments wherein X is Cl, the reaction is carried out using a chlorination reagent selected from the group consisting of 1,3-dichloro-5,5-dimethylhydantoin (DCDMH), N-chlorosuccinimide (NCS) and trichloroisocyanuric acid (TCCA).

In some embodiments the chlorination reagent is DCDMH.

In some embodiments the conversion of the compound of Formula (II) or a salt thereof to a compound of Formula (I) or a salt thereof is carried out in a suitable solvent and optionally with an additive material.

It was found that such a conversion was achieved in good yields with a number of solvents. In some embodiments the conversion of the compound of Formula (II) or a salt thereof

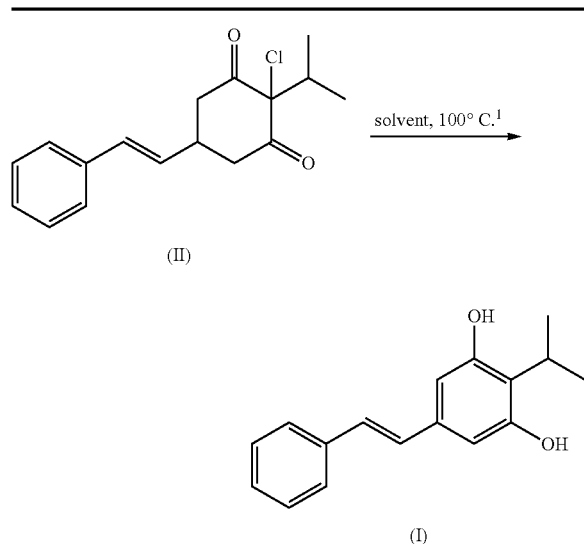
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to give a compound of Formula (I) or a salt thereof is carried out in a suitable solvent which is a polar aprotic solvent selected from the group consisting of dimethylformamide (DMF), 1,3-dimethyl-3,4,5,6-tetrahydro-2(1H)-pyrimidinone (DMPU), dimethylacetamide (DMAC) and sulfolane.

Whilst the use of a solvent such as DMF was operationally simple, the use of such in the manufacture of a pharmaceutical product is a matter of potential concern due to its reproductive toxicity. An alternative solvent was therefore sought that would be suitable for use in the large scale manufacture of a pharmaceutical product.

It was found, however, that the conversion of the compound of Formula (II) or a salt thereof, e.g. wherein X is Cl, to a compound of Formula (I) or a salt thereof did not progress well for alternative solvents, including some commonly used solvents. In such circumstances it was found that use of an additive material, such as benzyldiethylammonium chloride, facilitated this reaction in good yields. This is demonstrated in Table 1 below.

TABLE 1



Solvent	Yield without additive	Yield with additive ^{2,3}
Acetonitrile	No product	96%
Toluene	No product	98%
2-Me—THF	No product	95%
i-PrOAc	<5%	93%

¹ Reaction were carried out in sealed tubes which allowed for heating well above the boiling point of the solvent.

² Additive=2 equivalents of benzyldiethylammonium chloride.

³ Yields based on HPLC assay of crude reaction mixture.

A further investigation was therefore carried out into the use of a range of different additive materials in the conversion of the compound of Formula (II) or a salt thereof, in which X is Cl, to a compound of Formula (I) or a salt or solvate thereof. These results are represented in Table 2 below.

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TABLE 2

(II)

additive (2 equiv.)
acetonitrile
85° C.

(I)

Additive	Time	Yield ¹
BnEt ₃ Cl	4 hours	96%
Bu ₄ NCl	4 hours	93%
Et ₄ NCl	4 hours	99%
Me ₄ NCl	Overnight	90%
cysteine•HCl	Overnight	8%
NH ₄ Cl	Overnight	Trace
imidazole•HCl	5 hours	20%
Et ₃ N•HCl	Overnight	88%
Conc. HCl	5 hours	31%
NaCl	Overnight	Trace
LiCl	Overnight	20%

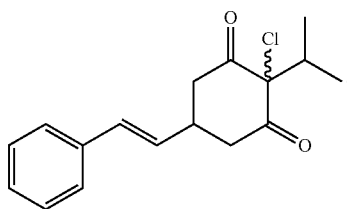
¹Yields based on HPLC assay of crude reaction mixture.

In some embodiments the conversion of the compound of Formula (II) or a salt thereof to a compound of Formula (I) or a salt thereof is carried out in the presence of an additive reagent which is a quaternary ammonium salt, for example a quaternary ammonium bromide salt or a quaternary ammonium chloride salt. In some embodiments the quaternary ammonium bromide salt is tetrabutylammonium bromide. In some embodiments the quaternary ammonium chloride salt is selected from the group consisting of benzyldiethylammonium chloride, tetrabutylammonium chloride, tetraethylammonium chloride and tetramethylammonium chloride. In some embodiments the quaternary ammonium chloride salt is tetraethylammonium chloride.

In some embodiments the conversion of the compound of Formula (II) or a salt thereof to give a compound of Formula (I) or a salt or a solvate thereof in the presence of an additive reagent is carried out in a solvent selected from the group consisting of acetonitrile, toluene, 2-methyl tetrahydrofuran, isopropyl acetate, acetone and methyl isobutyl ketone. It was found that acetonitrile provided the best combination of solubility and high boiling point. In some embodiments this conversion is carried out in a solvent which is acetonitrile.

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Embodiments of the present disclosure describe a compound of Formula (IIa) or a salt thereof



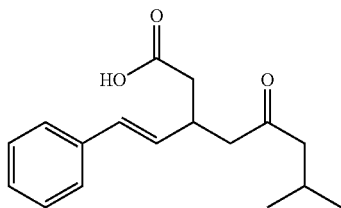
(IIa) 5

The compound of Formula (I) may be prepared in the form of an acetic acid solvate thereof. Some embodiments of the present disclosure describe a compound of Formula (I) in the form of an acetic acid solvate thereof. It has been found that formation of the acetic acid solvate of the compound of Formula (I) provides impurity and colour purging capabilities to the process of the invention. The acetic acid solvate of the compound of Formula (I) can thereafter be converted into the compound of Formula (I).

In some embodiments there is provided an acetic acid solvate of the compound of Formula (I) in crystalline solid state form. In some embodiments, there is provided, an acetic acid solvate of the compound of Formula (I) in crystalline solid state form which has an X-ray powder diffraction pattern substantially as shown in FIG. 2. In another embodiment there is provided an acetic acid solvate of the compound of Formula (I) in crystalline solid state characterised by an X-ray powder diffraction (XRPD) pattern with specific peaks at 6.7, 10.2, 11.1, 15.4, 16.9, 17.2, and 24.8 degrees (all 2θ values, ±0.1° 2θ experimental error). In another embodiment there is provided an acetic acid solvate of the compound of Formula (I) in crystalline solid state characterised by an X-ray powder diffraction (XRPD) pattern with at least six, or at least five, or at least four, or at least three specific peaks selected from peaks at 6.7, 10.2, 11.1, 15.4, 16.9, 17.2, and 24.8 degrees (all 2θ values, 0.1° 2θ experimental error).

In some embodiments the process further comprises a step of recrystallization of the compound of Formula (I) or a salt or solvate thereof. In some embodiments recrystallization is carried out using methanol and water.

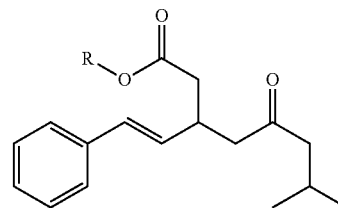
Process Step (b)
In some embodiments conversion of a compound of Formula (VI) or a salt thereof into a compound of Formula (III) or a salt thereof comprises a decarboxylation of the compound of Formula (VI) or a salt thereof to form the compound of Formula (V) or a salt thereof



(V) 55

followed by esterification of the compound of Formula (V) or a salt thereof to form the compound of Formula (IV) or a salt thereof

10



(IV)

wherein R is C₁₋₄ alkyl; and thereafter cyclization of the compound of Formula (IV) or a salt thereof to form the compound of Formula (III) or a salt thereof.

In some embodiments conversion of a compound of Formula (VI) or a salt thereof into a compound of Formula (V) or a salt thereof comprises decarboxylation in the presence of a base. In some embodiments the base is selected from imidazole, pyridine, and lutidine (2,6-dimethylpyridine). In a particular embodiment the base is triethylamine.

In some embodiments R is methyl, ethyl, propyl, or butyl. In some embodiments R is t-butyl. In some embodiments R is methyl.

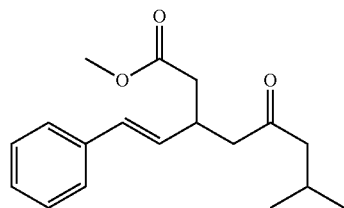
In some embodiments esterification of a compound of Formula (V) or a salt thereof to a compound of Formula (IV) or a salt thereof is performed using methanol, for example esterification is performed using methanol and hydrochloric acid.

In some embodiments cyclization of a compound of Formula (IV) or a salt thereof to a compound of Formula (III) or a salt thereof is performed using potassium tert-butoxide.

In some embodiments the compound of Formula (III) or a salt thereof is acidified and isolated by precipitation with methylcyclohexane.

In some embodiments the conversion of a compound of Formula (VI) or a salt thereof into a compound of Formula (III) or a salt thereof is telescoped such that the compounds of Formula (V) or a salt thereof and Formula (IV) or a salt thereof are not isolated.

Some embodiments describe a compound of Formula (IVa) or a salt thereof



(IVa)

The compound of Formula (IVa), namely (E)-methyl 7-methyl-5-oxo-3-styryloctanoate, is characterized by the following data.

¹H NMR (500 MHz, DMSO) δ 7.33 (m, 2H), 7.29 (m, 2H), 7.20 (m, 1H), 6.37 (d, 1H, J=16.0 Hz), 6.15 (dd, 1H, J=8.1 Hz, J=16.0 Hz), 3.56 (s, 3H), 3.09 (m, 1H), 2.60 (m, 2H), 2.50 (dd, 1H, 6.1 Hz, J=15.4 Hz), 2.42 (dd, 1H, J=8.1 Hz, J=15.4 Hz), 2.29 (d, 2H, 7.0 Hz), 2.00 (m, 1H), 0.82 (s, 6H, 6.7 Hz).

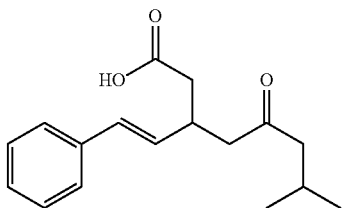
¹³C NMR (125 MHz, DMSO) δ 208.7 (C), 171.8 (C), 136.8 (C), 131.9 (CH), 129.5 (CH), 128.5 (CH), 127.2 (CH),

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125.9 (CH), 51.4 (CH₂), 51.2 (CH₃), 46.7 (CH₂), 38.5 (CH₂), 34.4 (CH), 23.8 (CH), 22.29 (CH₃), 22.26 (CH₃).

HRMS-APCI (m/z) [M+H]⁺ calculated for C₁₈H₂₅O₃, 289.1798; found, 289.1719.

In some embodiments there is provided a compound of Formula (V) or a salt thereof



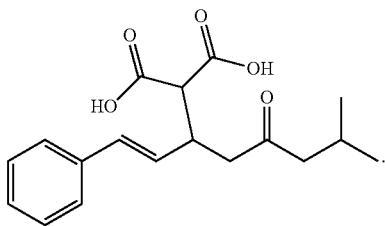
The compound of Formula (V), namely (E)-7-methyl-5-oxo-3-styryloctanoic acid, is characterized by the following data.

¹H NMR (700 MHz, DMSO) δ 12.13 (s, 1H), 7.33 (m, 2H), 7.29 (m, 2H), 7.20 (m, 1H), 6.37 (d, 1H, J=16 Hz), 6.17 (dd, 1H, J=8.0 Hz, J=16.0 Hz), 3.07 (m, 1H), 2.60 (m, 2H), 2.46 (dd, 1H, J=6.2 Hz, J=15.5 Hz), 2.33 (dd, 1H, J=8.0, J=15.5 Hz), 2.30 (d, 2H, J=7.1 Hz), 2.0 (m, 1H), 0.82 (d, 6H, J=6.6 Hz).

¹³C NMR (176 MHz, DMSO) δ 208.8 (C), 172.9 (C), 136.9 (C), 132.2 (CH), 129.3 (CH), 128.5 (CH), 127.1 (CH), 125.9 (CH), 51.4 (CH₂), 46.8 (CH₂), 38.9 (CH₂), 34.4 (CH), 23.8 (CH), 22.3 (CH₃), 22.3 (CH₃).

HRMS-APCI (m/z) [M+H]⁺ calcd for C₁₇H₂₃O₃, 275.1642; found, 275.1635.

In some embodiments there is provided a compound of Formula (VI) or a salt thereof.

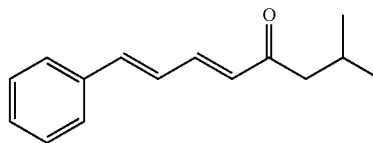


The compound of Formula (III) or a salt thereof may be converted into a compound of Formula (I) or a salt or solvate thereof by methods described herein (e.g. process step (a)) or by methods known to the person skilled in the art (e.g. by methods described in Kronenwerth, M. et al.). In some embodiments the compound of Formula (III) or a salt thereof may be converted into a compound of Formula (I) or a salt or solvate thereof by process step (a).

Process Step (c)

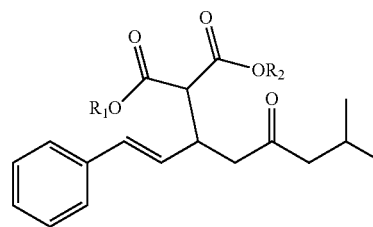
In some embodiments conversion of a compound of Formula (IX) or a salt thereof into a compound of Formula (VI) or a salt thereof comprises the condensation of the compound of Formula (IX) or a salt thereof with methyl isobutyl ketone to form the compound of Formula (VIII) or a salt thereof

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(VIII)

followed by addition of a malonic ester (R₁O(O)C—CH₂—C(O)OR₂) to form a compound of Formula (VII) or a salt thereof



(VII)

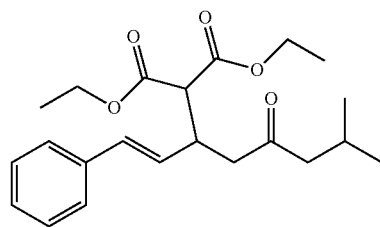
wherein R₁ and R₂ are independently C₁₋₄ alkyl; and thereafter hydrolysis of the compound of Formula (VII) or a salt thereof to form the compound of Formula (VI) or a salt thereof.

In some embodiments condensation of a compound of Formula (IX) or a salt thereof into a compound of Formula (VIII) or a salt thereof is performed using lithium, potassium or sodium hydroxide, for example the condensation is performed using potassium or sodium hydroxide in methanol.

In some embodiments the conversion of a compound of Formula (VIII) or a salt thereof into a compound of Formula (VII) or a salt thereof comprises the addition of a malonic ester which is di-tert-butyl malonate or diethyl malonate.

In some embodiments the conversion of a compound of Formula (IX) or a salt thereof into a compound of Formula (VI) or a salt thereof is telescoped such that the compounds of Formula (VII) and Formula (VIII) or salts thereof are not isolated.

In some embodiments there is provided a compound of Formula (VIIa) or a salt thereof



(VIIa)

The compound of Formula (VIIa), namely (E)-diethyl 2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonate, is characterized by the following data.

¹H NMR (700 MHz, DMSO) δ 7.30 (m, 2H), 7.30 (m, 2H), 7.21 (m, 1H), 6.39 (d, 1H, J=15.8 Hz), 6.14 (dd, 1H, J=15.99 Hz, J=8.9 Hz), 4.12 (q, 2H, J=7.13 Hz), 4.06 (m, 2H), 3.68 (d, 1H, J=8.1 Hz), 3.33 (m, 1H), 2.73 (dd, 1H, J=16.9 Hz, J=9.0 Hz), 2.63 (m, 1H), 2.28 (d, 2H, 6.9 Hz), 1.98 (m, 1H, 6.7 Hz), 1.16 (t, 3H, J=7.1 Hz), 1.10 (t, 3H, J=7.0 Hz), 0.81 (d, 3H, J=6.6 Hz), 0.80 (d, 3H, J=6.7 Hz).

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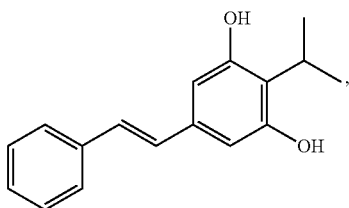
^{13}C NMR (176 MHz, DMSO) δ 208.1 (C), 167.7 (C), 167.6 (C), 136.6 (C), 131.4 (CH), 128.9 (CH), 127.4 (CH), 125.9 (CH), 61.0 (CH₂), 60.8 (CH₂), 55.0 (CH), 51.3 (CH₂), 44.8 (CH₂), 37.7 (CH), 23.8 (CH), 22.3 (CH₃), 22.2 (CH₃), 13.9 (CH₃), 13.9 (CH₃).

HRMS-APCI (m/z) [M+H]⁺ calcd for C₂₂H₃₁O₅, 375.2166; found, 375.2158.

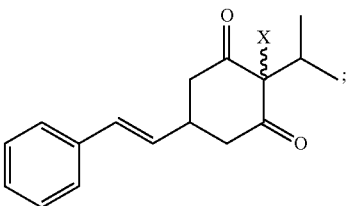
The compound of Formula (VI) or a salt thereof may be converted into a compound of Formula (I) or a salt or solvate thereof by methods described herein (e.g. process steps (a) and (b)) or by methods known to the person skilled in the art (e.g. by methods described in Kronenwerth, M. et al.). In some embodiments the compound of Formula (VI) or a salt thereof may be converted into a compound of Formula (I) or a salt or solvate thereof by process steps (a) and (b).

Preparation of the Compound of Formula (I)-Process B

Some embodiments herein describe a process for preparing a compound of Formula (I) or salt or solvate thereof



comprising aromatizing a compound of Formula (II) or salt thereof

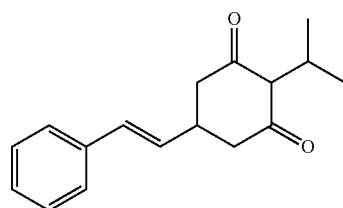


wherein X is Cl, Br, or I; to obtain the compound of Formula (I). In some embodiments the process further comprises purifying the compound of Formula (I) or a salt or solvate thereof. In some embodiments the purifying comprises crystallization of the compound of Formula (I) or a salt or solvate thereof. In some embodiments the compound of Formula (I) is a crystal form described in any embodiment described herein. In some embodiments the compound of Formula (I) is crystal form 1. In some embodiments the compound of Formula (I) is an anhydrous crystal. In some embodiments the compound of Formula (I) has an X-ray powder diffraction pattern substantially as shown in FIG. 1. In some embodiments the compound of Formula (I) is in a crystalline solid state characterised by an X-ray powder diffraction (XRPD) pattern with specific peaks at 15.0, 17.8, 19.1, 20.2, 21.5, 22.4, 23.3, 24.5, 26.2 and 27.9 degrees (all values $\pm 0.1^\circ$ 2θ experimental error). In some embodiments the compound of Formula (I) is in a crystalline solid state

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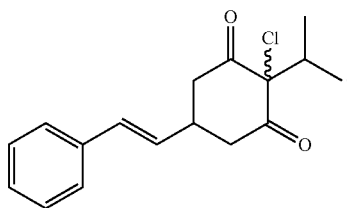
characterised by an X-ray powder diffraction (XRPD) pattern with at least nine, or at least eight, or at least seven, or at least six, or at least five, or at least four, or at least three specific peaks selected from 15.0, 17.8, 19.1, 20.2, 21.5, 22.4, 23.3, 24.5, 26.2 and 27.9 degrees (all values $\pm 0.1^\circ$ 2θ experimental error). In some embodiments the compound of Formula (I) is an acetic acid solvate in crystalline solid state form. In some embodiments, the compound of Formula (I) is an acetic acid solvate in crystalline solid state form which has an X-ray powder diffraction pattern substantially as shown in FIG. 2. In another embodiment the compound of Formula (I) is an acetic acid solvate in crystalline solid state characterised by an X-ray powder diffraction (XRPD) pattern with specific peaks at 6.7, 10.2, 11.1, 15.4, 16.9, 17.2, and 24.8 degrees (all 2θ values, $\pm 0.1^\circ$ 2θ experimental error). In another embodiment the compound of Formula (I) is an acetic acid solvate in crystalline solid state characterised by an X-ray powder diffraction (XRPD) pattern with at least six, or at least five, or at least four, or at least three specific peaks selected from peaks at 6.7, 10.2, 11.1, 15.4, 16.9, 17.2, and 24.8 degrees (all 2θ values, $\pm 0.1^\circ$ 2θ experimental error). In some embodiments X is Cl. In some embodiments the aromatizing is carried out in a suitable solvent and optionally with an additive reagent. In some embodiments the aromatizing is carried out in a suitable solvent which is a polar aprotic solvent selected from the group consisting of dimethylformamide (DMF), 1,3-dimethyl-3,4,5,6-tetrahydro-2(1H)-pyrimidinone. In some embodiments the aromatizing is carried out in the presence of an additive reagent in a solvent selected from the group consisting of acetonitrile, toluene, 2-methyl tetrahydrofuran, isopropyl acetate, acetone and methyl isobutyl ketone. In some embodiments the aromatizing is carried out in acetonitrile in the presence of an additive reagent. In some embodiments the additive reagent is a quaternary ammonium salt. In some embodiments the quaternary ammonium salt is selected from the group consisting of tetrabutylammonium bromide, benzyltriethylammonium chloride, tetrabutylammonium chloride, tetraethylammonium chloride and tetramethylammonium chloride. In some embodiments the quaternary ammonium salt is tetraethylammonium chloride. In some embodiments the aromatizing is carried out in acetonitrile with tetraethylammonium chloride.

In some embodiments the process further comprises halogenating a compound of Formula (III) or salt thereof



with a halogenating agent to obtain the compound of Formula (II) or a salt thereof. In some embodiments the halogenating agent is selected from 1,3-dichloro-5,5-dimethylhydantoin (DCDMH); N-chlorosuccinimide (NCS); and trichloroisocyanuric acid (TCCA). In some embodiments the halogenating agent is DCDMH and the compound of Formula (II) or a salt thereof is a compound of Formula (IIa) or a salt thereof:

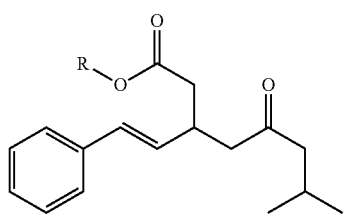
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(IIa)

In some embodiments the halogenating is carried out in methanol. In some embodiments the halogenation agent is DCDMH and the halogenation is carried out in methanol.

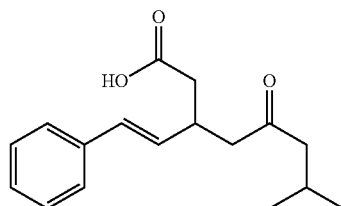
In some embodiments the process further comprises cyclizing a compound of Formula (IV) or salt thereof



(IV)

wherein R is C_{1-4} alkyl, to obtain the compound of Formula (III) or a salt thereof. In some embodiments R is selected from the group consisting of methyl, ethyl, propyl, or butyl. In some embodiments R is t-butyl. In some embodiments R is methyl. In some embodiments the cyclization comprises contacting the compound of Formula (IV) or a salt thereof, with a base. In some embodiments the cyclization is performed using potassium tert-butoxide. In some embodiments the cyclization is carried out in 2-methyltetrahydrofuran. In some embodiments the cyclization comprises treating a compound of Formula (IV) or a salt thereof, for example a compound of (IVa) or a salt thereof, with potassium tert-butoxide in 2-methyltetrahydrofuran. In some embodiments the compound of Formula (III) or a salt thereof is further acidified and isolated by precipitation with methylcyclohexane.

In some embodiments the process further comprises esterifying a compound of Formula (V) or salt thereof

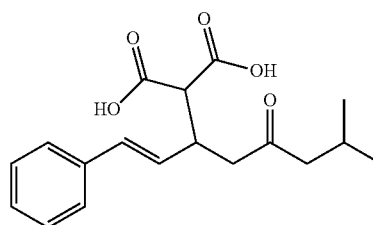


(V)

to obtain the compound of Formula (IV) or a salt thereof. In some embodiments the esterifying is carried out using methanol and hydrochloric acid to obtain compound (IV). In some embodiments the esterifying comprises treating the compound of Formula (V) or a salt thereof with aqueous hydrochloric acid in methanol.

In some embodiments the process further comprising decarboxylating a compound of Formula (VI) or a salt thereof

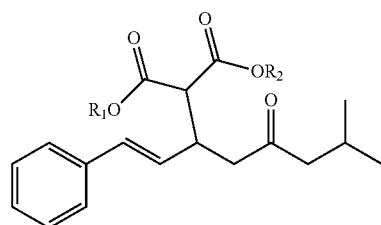
16



(VI)

to obtain the compound of Formula (V) or a salt thereof. In some embodiments the decarboxylating comprises the presence of a base. In some embodiments the base is selected from imidazole, pyridine, and lutidine (2,6-dimethylpyridine). In some embodiments the base is trimethylamine. In some embodiments the decarboxylating comprises heating the compound of Formula (VI) or a salt thereof in the presence of trimethylamine.

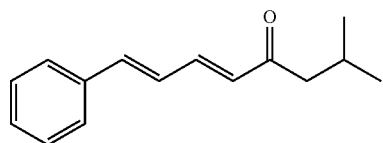
In some embodiments the process further comprises hydrolyzing a compound of Formula (VII) or a salt thereof



(VII)

wherein each R_1 and R_2 is independently C_{1-4} alkyl; to obtain the compound of Formula (VI) or a salt thereof. In some embodiments each of R_1 and R_2 is ethyl (VIIa). In some embodiments the hydrolyzing comprises treating the compound of Formula (VII) or a salt thereof, with sodium hydroxide. In some embodiments the hydrolyzing comprises treating the compound of Formula (VII) or a salt thereof with sodium hydroxide in ethanol. In some embodiments the process comprises hydrolyzing a compound of Formula (VIIa) or a salt thereof comprising treating the compound of Formula (VIIa) or a salt thereof with sodium hydroxide in ethanol.

In some embodiments the process further comprises adding a dialkyl malonic ester ($R_1O(O)C-CH_2-C(O)OR_2$ wherein each R_1 and R_2 is independently C_{1-4} alkyl), to a compound of Formula (VIII) or a salt thereof

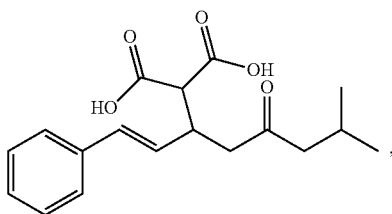


(VIII)

to obtain the compound of Formula (VII) or a salt thereof. In some embodiments the dialkyl malonic ester is di-tert-butylmalonate or diethyl malonate. In some embodiments, the adding comprises contacting the dialkyl malonic ester with the compound of Formula (VIII) or a salt thereof in the presence of lithium bromide/triethylamine.

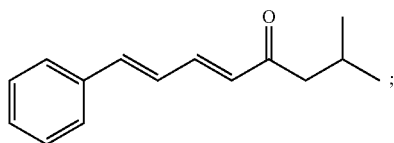
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in the presence of a base, to form the compound of Formula (V) or a salt thereof. Some embodiments describe the compound of Formula (IVa) or a salt thereof wherein the compound of Formula (V) or a salt thereof is prepared by any process described herein. Some embodiments describe the compound of Formula (IVa) or a salt thereof wherein the compound of Formula (V) or a salt thereof is prepared by a process comprising decarboxylating a compound of Formula (VI) or a salt thereof

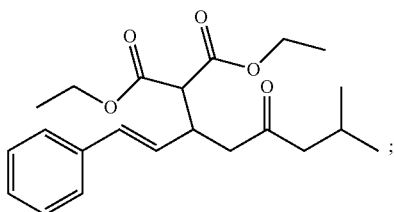


in the presence of a base, to form the compound of Formula (V) or a salt thereof. Some embodiments describe the compound of Formula (IVa) or a salt thereof wherein the compound of Formula (VI) or a salt thereof is prepared by any process described herein. Some embodiments describe the compound of Formula (IVa) or a salt thereof wherein the compound of Formula (VI) or a salt thereof is prepared by a process comprising:

- i. condensing trans-cinnamaldehyde or a salt thereof with methyl isobutyl ketone to form a compound of Formula (VIII) or a salt thereof



- ii. adding diethyl malonate to the compound of Formula (VIII) or a salt thereof to form a compound of Formula (VIIa) or a salt thereof

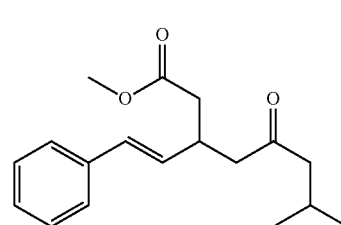


and

- iii. hydrolyzing the compound of Formula (VIIa) or a salt thereof to form the compound of Formula (VI) or a salt thereof.

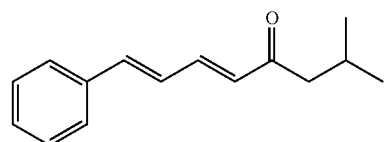
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Some embodiments herein describe a compound of Formula (IVa) or a salt thereof

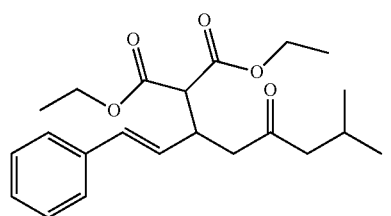


prepared by a process comprising:

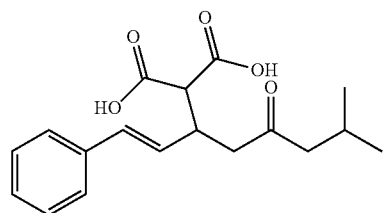
- a) condensing trans-cinnamaldehyde or a salt thereof with methyl isobutyl ketone to form a compound of Formula (VIII) or a salt thereof



- b) adding diethyl malonate to the compound of Formula (VIII) or a salt thereof to form a compound of Formula (VIIa) or a salt thereof



- c) hydrolyzing the compound of Formula (VIIa) or a salt thereof to form the compound of Formula (VI) or a salt thereof



63

Me methyl
 2-MeTHF 2-methyl tetrahydrofuran
 MHz megahertz
 MIBK methyl isobutyl ketone
 MS mass spectrometry
 NaOH sodium hydroxide
 NEt₃ triethylamine
 NH₄Cl ammonium chloride
 NMR Nuclear Magnetic Resonance
 TBAC tetrabutylammonium chloride
 TBME tert-butyl methyl ether
 tBu tert-butyl
 tBuOK potassium tert-butoxide
 TCCA trichloroisocyanuric acid
 TEAC tetraethylammonium chloride

EXAMPLES

General Experimental Procedures. All reactions were performed under positive pressure of nitrogen in a Jacketed Laboratory Reactor equipped with overhead stirring and fitted with Teflon septa.

Materials: Commercial solvents and reagents were used as received.

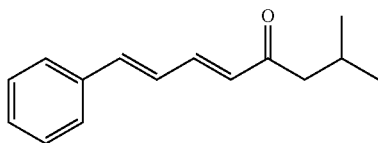
Instrumentation:

Unless otherwise stated proton nuclear magnetic resonance spectra (¹H NMR) were recorded at 400 MHz at 25° C. Chemical shifts are expressed in parts per million (ppm, 6 scale) downfield from tetramethylsilane and are referenced to residual proton in the NMR solvent. Data are represented as follows: chemical shift, multiplicity (s=singlet, d=doublet, sep=septet, m=multiplet and/or multiple resonances, br=broad), integration, coupling constant in Hertz, and assignment. Proton-decoupled carbon nuclear magnetic resonance spectra (¹³C NMR) were recorded at 100 MHz at 25° C. Chemical shifts are expressed in parts per million (ppm, δ scale) downfield from tetramethylsilane and are referenced to the carbon resonances of the solvent. High-resolution mass spectrometry (IRMS) were obtained using an Orbitrap mass analyzer.

The X-ray powder diffraction (XRPD) data were acquired on a PANalytical X'Pert Pro powder diffractometer, model PW3050/60, using an X'Celerator detector. The acquisition conditions were: radiation: Cu Kα, generator tension: 45 kV, generator current: 40 mA, step size: 0.017°2θ, time per step: 500 seconds, divergence slit type: fixed, divergence slit size: 0.4354°, measurement temperature: 20-25° C., goniometer radius: 240 mm. The sample was prepared by packing sample in a 0.9 mm capillary, or zero background silicone sample holder. Peak positions were obtained using PANalytical X'Pert Highscore Plus software. The margin of error is approximately ±0.1° 2θ for each of the peak assignments.

Example 1

Synthesis of (5E,7E)-2-methyl-8-phenylocta-5,7-dien-4-one, (Compound of Formula (VIII))

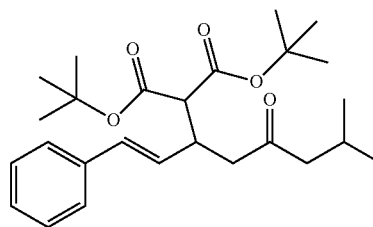


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A solution of sodium hydroxide (11.4 g, 284 mmol, 1.5 equiv) in methanol (100 mL) was added dropwise over 1 hour to a stirred solution of cinnamaldehyde (compound of Formula (IX)) (25 g, 189 mmol, 1 equiv) and methyl isobutyl ketone (75 mL, 602 mmol, 3.2 equiv) at -5° C. Upon completion of the addition, the reaction mixture was stirred for 2 hours at 0° C. Toluene (250 mL) and water (125 mL) were then added directly to the reactor at 0° C. and the stirring biphasic mixture allowed to warm to 20° C. The aqueous layer was removed and the organic layer was washed with water (2×75 mL). The washed organic layer was then dried via azeotropic distillation under vacuum to a final solution volume of 75 mL or until analysis via Karl Fischer titration revealed water content less than 0.2%. The crude material was utilized in the next transformation without any further processing.

Example 2

Synthesis of (E)-di-tert-butyl 2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonate, (Compound of Formula VII in which R₁+R₂=tBu)



(VII)

The concentrated solution obtained in the previous step (example 1) was diluted with toluene such that the total volume of the solution was 175 mL. To the solution of (5E,7E)-2-methyl-8-phenylocta-5,7-dien-4-one, (compound of Formula (VIII)) in toluene at 20° C. was then added di-tert-butyl malonate (40.3 mL, 180 mmol, 0.95 equiv), triethylamine (13.1 mL, 94.5 mmol, 0.5 equiv), and powdered lithium bromide (3.28 g, 37.8 mmol, 0.2 equiv). The heterogeneous reaction mixture was stirred for 2 hours at 20° C. Water (200 mL) was then charged directly to the stirring reaction mixture. The layers were allowed to separate and the aqueous layer discarded. The resulting product solution was utilized in the next transformation without any further processing. Alternatively, the di-tert-butyl malonate addition product (E)-di-tert-butyl 2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonate (compound of Formula (VII)) could also be isolated via concentration of the toluene solution and crystallization from isopropyl alcohol/water (3:2) to provide (E)-di-tert-butyl 2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonate (compound of Formula (VII)) as a light yellow solid.

(VIII) 60

¹H NMR (400 MHz, DMSO) δ 7.32-7.29 (m, 4H), 7.24-7.18 (m, 1H), 6.38 (d, 1H, J=16 Hz), 6.12 (dd, 1H, J=16, 8.4 Hz), 3.39 (d, 1H, J=8.4 Hz), 3.27-3.20 (m, 1H), 2.70 (dd, 1H, J=16.8, 9.2 Hz), 2.58 (dd, 1H, J=16.8, 4.4 Hz), 2.28 (d, 2H, J=7.2 Hz), 1.98 (sep, 1H, J=6.8 Hz), 1.39 (s, 9H), 1.33 (s, 9H), 0.81 (d, 3H, J=6.4 Hz), 0.80 (d, 3H, J=6.4 Hz).

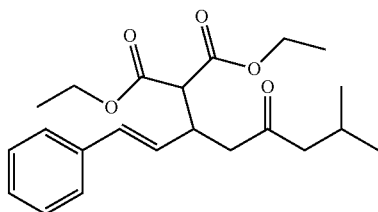
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^{13}C NMR (100 MHz, DMSO) δ 208.2 (C), 167.0 (C), 166.8 (C), 136.7 (C), 131.2 (CH), 129.1 (CH), 128.6 (CH), 127.3 (CH), 125.9 (CH), 81.2 (C), 80.9 (C), 56.8 (CH), 51.4 (CH₂), 45.0 (CH₂), 37.7 (CH), 27.5 (CH₃), 27.4 (CH₃), 23.9 (CH), 22.3 (CH₃), 22.2 (CH₃).

HRMS-APCI (m/z) [M+H]⁺ calcd for C₂₆H₃₉O₅, 431.2792; found, 431.2754.

Example 3

Synthesis of (E)-diethyl 2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonate, (Compound of Formula VIIa)



A concentrated solution of (5E,7E)-2-methyl-8-phenylocta-5,7-dien-4-one (about 7.6 moles) is diluted with toluene. To the solution of (5E,7E)-2-methyl-8-phenylocta-5,7-dien-4-one, (compound of Formula (VIII)) in toluene at 20° C. is then added diethyl malonate (1.83 kg, 1.73 L, 1.5 equiv), triethylamine (1.15 kg, 1.58 L, 1.5 equiv), and powdered lithium bromide (131 g, 0.2 equiv). The heterogeneous reaction mixture is stirred at 35-40° C. for at least 8 hours. Water (8 L, 8 vols) is then charged directly to the stirring reaction mixture and stirred at 35° C. for 15-30 minutes. The layers are allowed to separate and the aqueous layer discarded. The resulting product solution is utilized in the next transformation without any further processing. Alternatively, the diethyl malonate addition product (E)-diethyl 2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonate (compound of Formula (VIIa)) can also be isolated via concentration of the toluene solution and crystallization to provide (E)-diethyl 2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonate (compound of Formula (VIIa)).

^1H NMR (700 MHz, DMSO) δ 7.30 (m, 2H), 7.30 (m, SH₀), 7.21 (m, 1H), 6.39 (d, 1H, J=15.8 Hz), 6.14 (dd, 1H, J=15.99 Hz, J=8.9 Hz), 4.12 (q, 2H, J=7.13 Hz), 4.06 (m, 2H), 3.68 (d, 1H, J=8.1 Hz), 3.33 (m, 1H), 2.73 (dd, 1H, J=16.9 Hz, J=9.0 Hz), 2.63 (m, 1H), 2.28 (d, 2H, 6.9 Hz), 1.98 (m, 1H, 6.7 Hz), 1.16 (t, 3H, J=7.1 Hz), 1.10 (t, 3H, J=7.0 Hz), 0.81 (d, 3H, J=6.6 Hz), 0.80 (d, 3H, J=6.7 Hz).

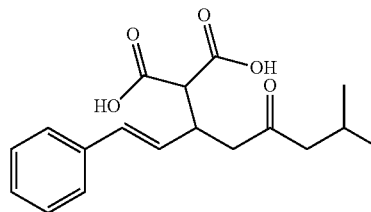
^{13}C NMR (176 MHz, DMSO) δ 208.1 (C), 167.7 (C), 167.6 (C), 136.6 (C), 131.4 (CH), 128.9 (CH), 127.4 (CH), 125.9 (CH), 61.0 (CH₂), 60.8 (CH₂), 55.0 (CH), 51.3 (CH₂), 44.8 (CH₂), 37.7 (CH), 23.8 (CH), 22.3 (CH₃), 22.2 (CH₃), 13.9 (CH₃), 13.9 (CH₃).

HRMS-APCI (m/z) [M+H]⁺ calcd for C₂₂H₃₁O₅, 375.2166; found, 375.2158.

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Example 4

Synthesis of (E)-2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonic Acid, (Compound of Formula (VI))



Method A

The toluene solution of (E)-di-tert-butyl 2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonate (compound of Formula (VII)) obtained in the previous step was diluted with acetic acid (50 mL) and added drop wise over 1 hour to a stirred solution consisting of acetic acid (100 mL) and concentrated aqueous hydrochloric acid (75 mL) at 60° C. Upon completion of the addition, the resulting solution was stirred for 4 hours at 60° C. The product mixture was allowed to cool over 30 minutes to 20° C. and water (200 mL) added. The two layers were stirred vigorously at 20° C. and then allowed to settle. The aqueous layer was then discarded and the toluene layer concentrated to dryness. To the resulting oil was added toluene (250 mL) and the solution heated to 60° C. with stirring. The warm toluene solution was then allowed to cool slowly over 1 hour to 20° C. at which point (E)-2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonic acid (compound of Formula (VI)) precipitated from solution. The solid was then filtered and the wet cake washed with toluene (100 mL). The wet cake was then dried under vacuum at 30° C. for 12 hours to afford (E)-2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonic acid (compound of Formula (VI)) as a white crystalline solid (37.3 g, 62% from cinnamaldehyde (compound of Formula (IX)).

Method B

A toluene solution of (E)-diethyl 2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonate from example 3 (about 7.6 moles, compound of Formula (VIIa)) at 20-30° C. was charged with 6M aqueous sodium hydroxide (5.05 L, 4 equiv), toluene (0.5 L or 0.5 vols), and 200 proof ethanol (2 vols) and the contents were stirred for at least 4 h at T_f=20° C. Once complete, the temperature was adjusted to 35-50° C. and after stirring for at least 30 minutes agitation was stopped and the layers were separated. The aqueous layer was cooled to 0-5° C. and the pH was adjusted with conc. HCl to a pH of 0-1 (required 2.9 L or 4.7 equiv), while maintaining the temperature at 10° C. or less during the addition. Once the desired pH was reached the vessel was charged with TBME (3 L, 3 vols) and the biphasic mixture was warmed to 20-25° C. The mixture was stirred for 15-30 minutes and the layers were separated. To the organic layer was added toluene (7 L, 7 vols) and water (6 L, 6 vols) and the mixture was stirred for 15-30 minutes. The layers were separated and toluene was added to achieve a fill of 16-18 vols (3 L, 3 vols). The mixture was distilled down to about 9-9.5 vols using vacuum distillation, the temperature was adjusted to 40-45° C. and the mixture was seeded with 5 g of compound VI (0.2% w/w against theoretical yield). The mixture was stirred at 40-45° C. for 30-60 minutes. Once

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nucleation was observed the mixture was diluted to 12 volumes with toluene and held at 40-45° C. for at least 1 hour. The slurry was cooled to 10-20° C. at 0.5° C./min and held at 10-20° C. for at least 1 hour. The solid was isolated by filtration and the filter cake was washed with toluene (2x 7 vols). The solid was dried at 25-35° C. overnight in a vacuum oven to afford (E)-2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonic acid (compound of Formula (VI))

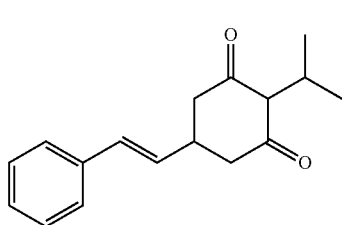
¹H NMR (400 MHz, DMSO) δ 12.78 (br s, 2H), 7.32-7.27 (m, 4H), 7.23-7.19 (m, 1H), 6.39 (d, 1H, J=16 Hz), 6.17 (dd, 1H, J=16, 8.4 Hz), 3.43 (d, 1H, J=8.4 Hz), 3.31-3.24 (m, 1H), 2.72 (dd, 1H, J=16.8, 8.8 Hz), 2.63 (dd, 1H, J=16.4, 4 Hz), 2.28 (d, 2H, J=7.2 Hz), 1.98 (sep, 1H, J=6.8 Hz), 0.81 (d, 3H, J=6.4 Hz), 0.80 (d, 3H, J=6.4 Hz).

¹³C NMR (100 MHz, DMSO) δ 208.4 (C), 169.6 (C), 169.5 (C), 136.8 (C), 130.9 (CH), 129.6 (CH), 128.5 (CH), 127.3 (CH), 126.0 (CH), 55.7 (CH), 51.4 (CH₂), 45.1 (CH₂), 37.6 (CH), 23.9 (CH), 22.3 (CH₃), 22.2 (CH₃).

HRMS-APCI (m/z) [M+H]⁺ calcd for C₁₈H₂₃O₅, 319.1540; found, 319.1528.

Example 5

Synthesis of (E)-2-isopropyl-5-styrylcyclohexane-1,3-dione, (Compound of Formula (III))



Triethylamine (4.4 mL, 31.4 mmol, 0.25 equiv) was charged to a stirred slurry of (E)-2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonic acid (compound of Formula (VI)) (40.0 g, 126 mmol, 1 equiv) in toluene (210 mL). The resulting mixture was heated to 110° C. and allowed to stir 2 hours. The reaction mixture containing the non-isolated intermediate (E)-7-methyl-5-oxo-3-styryloctanoic acid (compound of Formula V) was then allowed to cool to 20° C. and charged with methanol (120 mL) and concentrated aqueous hydrochloric acid (10.5 mL). The resulting solution was stirred for 4 hours at 60° C. The resulting mixture, containing the non-isolated intermediate (E)-methyl 7-methyl-5-oxo-3-styryloctanoate (compound of Formula (IVa)), was allowed to cool to 20° C. and washed with water (200 mL). The washed organic layer was then dried via vacuum distillation to a final solution volume of 120 mL. The solution of the methyl ester (E)-methyl 7-methyl-5-oxo-3-styryloctanoate (compound of Formula (IVa)) was then cooled to 0° C. A solution of potassium tert-butoxide (19.7 g, 176 mmol, 1.4 equiv) in 2-methyltetrahydrofuran (80 mL) was then added drop wise over 1 hour to the cooled solution of (E)-methyl 7-methyl-5-oxo-3-styryloctanoate (compound of Formula (IVa)). Upon completion of the addition, the reaction mixture was warmed over 30 minutes to 20° C. The warmed solution was stirred 1 hour at 20° C. The resulting product mixture was then charged with 1.0 M aqueous hydrochloric acid (180 mL) and the biphasic mixture stirred 10 minutes. The aqueous layer was then discarded and the organic layer washed with 10% aqueous sodium chloride

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(2x160 mL). To the solution of the (E)-2-isopropyl-5-styrylcyclohexane-1,3-dione (compound of Formula (III)) was then added methylcyclohexane (400 mL) drop wise over 1 hour at which point the product began to precipitate out. The slurry was then filtered and the wet cake washed with methylcyclohexane (120 mL). The wet cake was then dried under vacuum at 30° C. for 12 hours to provide (E)-2-isopropyl-5-styrylcyclohexane-1,3-dione (compound of Formula (III)) as a white crystalline solid (25.3 g, 79% from (E)-2-(7-methyl-5-oxo-1-phenyloct-1-en-3-yl)malonic acid, (compound of Formula (VI)).

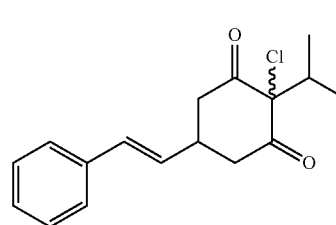
¹H NMR (400 MHz, DMSO) δ 10.33 (br s, 1H, OH), 7.39-7.37 (m, 2H), 7.34-7.29 (m, 2H), 7.24-7.20 (m, 1H), 6.43 (d, 1H, J=16 Hz), 6.27 (dd, 1H, J=16, 7.2 Hz), 3.08 (sep, 1H, J=7.2 Hz), 2.87-2.78 (m, 1H), 2.40 (br, 4H), 1.08 (d, 6H, J=7.2 Hz).

¹³C NMR (100 MHz, DMSO) δ 136.8 (C), 132.5 (CH), 128.9 (CH), 128.6 (CH), 127.3 (CH), 126.0 (CH), 119.2 (C), 36.1 (CH), 22.6 (CH), 20.4 (CH₃), other carbons undergo enol tautomerization.

HRMS-APCI (m/z) [M+H]⁺ calcd for C₁₇H₂₁O₂, 257.1536; found, 257.1519.

Example 6

Synthesis of (E)-2-chloro-2-isopropyl-5-styrylcyclohexane-1,3-dione, (Compound of Formula (IIa))



A 1-L jacketed laboratory reactor was charged with (E)-2-isopropyl-5-styrylcyclohexane-1,3-dione (compound of Formula (III)) (55 g, 215 mmol, 1 equiv) and methanol (495 mL) resulting in a heterogeneous suspension. The mixture was heated to an internal temperature of ~45-50° C. at which point a homogeneous solution was achieved. To this solution was charged solid 1,3-dichloro-5,5-dimethylhydantoin (23.3 g, 118 mmol, 0.55 equiv) sequentially, in five equal portions. The product solution was then cooled to 40° C. over 10 minutes and seeded with 146 mg of (E)-2-chloro-2-isopropyl-5-styrylcyclohexane-1,3-dione (compound of Formula (IIa)). The mixture was aged for 30 minutes then cooled to 23° C. over 60 minutes. Water (330 mL) was then added drop wise to the slurry over 1 hour and the mixture stirred 30 minutes at 20° C. The slurry was then filtered and the wet cake washed with 1:1 methanol/water (2x110 mL). The wet cake was then dried under vacuum at 45° C. for 12 hours to provide (E)-2-chloro-2-isopropyl-5-styrylcyclohexane-1,3-dione (compound of Formula (IIa)) as a white crystalline solid existing as a 2:1 mixture of diastereomers (60.0 grams, 96%).

Isomer 1:

¹H NMR (400 MHz, DMSO) δ 7.41-7.39 (m, 2H), 7.36-7.32 (m, 2H), 7.23-7.22 (m, 1H), 6.54 (d, 1H, J=16 Hz), 6.32 (dd, 1H, J=16, 7.2 Hz), 3.20-3.14 (m, 3H), 2.84-2.77 (m, 1H), 2.73-2.72 (m, 1H), 2.70-2.69 (m, 1H), 0.82 (d, 6H, J=6.8 Hz).

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¹³C NMR (100 MHz, DMSO) δ 198.5 (C), 136.5 (C), 130.4 (CH), 129.9 (CH), 128.6 (CH), 127.5 (CH), 126.1 (CH), 94.6 (C), 43.4 (CH₂), 35.0 (CH), 32.8 (CH), 16.7 (CH₃).

Isomer 2:

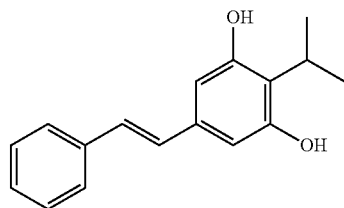
¹H NMR (400 MHz, DMSO) δ 7.33-7.28 (m, 4H), 7.25-7.21 (m, 2H), 6.35 (dd, 1H, J=16, 2 Hz), 6.03 (dd, 1H, J=16, 5.6 Hz), 3.48 (dd, 1H, J=14.8, 6 Hz), 3.24-3.18 (m, 1H), 3.10 (sep, 1H, J=6.4 Hz), 2.86 (dd, 1H, J=14.8, 3.6 Hz), 0.83 (d, 6H, J=6.4 Hz).

¹³C NMR (100 MHz, DMSO) δ 199.6 (C), 136.1 (C), 130.3 (CH), 130.1 (CH), 128.6 (CH), 127.7 (CH), 126.1 (CH), 94.5 (C), 42.2 (CH₂), 34.7 (CH), 30.9 (CH), 16.9 (CH₃).

HRMS-APCI (m/z) [M+H]⁺ calcd for C₁₇H₂₀ClO₂, 291.1146; found, 291.1132.

Example 7

Synthesis of (E)-2-isopropyl-5-styrylbenzene-1,3-diol, (Compound of Formula (I))



A jacketed laboratory reactor was charged with (E)-2-chloro-2-isopropyl-5-styrylcyclohexane-1,3-dione (compound of Formula (IIa)) (407 g, 1.4 mol, 1 equiv), tetraethylammonium chloride (464 g, 2.8 mol, 2 equiv), and acetonitrile (1.2 L). The mixture was heated to 75-80° C. for 6 hours. The product mixture was then cooled over 30 minutes to 20° C. and diluted with tert-butyl methyl ether (3.26 L). The resulting solution was washed four times with water (2 L each wash). The resulting solution was concentrated to a volume of ~610 ml. Acetic acid (815 ml) was charged and the solution was concentrated under reduced pressure to a volume of ~1.3 L. The resultant slurry was heated to 55° C. until homogeneous then cooled slowly to 35° C. over the course of 1 hour during which time product precipitation commenced. Methylcyclohexane (6.5 L) was added to the slurry over the course of 2 hours. Once the addition was complete, the mixture was further cooled to ~23° C. over one hour. The solids were filtered and washed twice with 6:1 methylcyclohexane/AcOH. The isolated solid was then dried under vacuum at 80° C. for 24 hours to provide (E)-2-isopropyl-5-styrylbenzene-1,3-diol (compound of Formula (I)) as a white crystalline solid (305 g, 86%).

¹H NMR (400 MHz, DMSO) δ 9.05 (s, 2H, OH), 7.58-7.55 (m, 2H), 7.37-7.33 (m, 2H), 7.27-7.22 (m, 1H), 7.00 (d, 1H, J=16.4 Hz), 6.87 (d, 1H, J=16.4 Hz), 6.47 (s, 2H), 3.43 (sep, 1H, J=7.2 Hz), 1.24 (d, 6H, J=7.2 Hz).

¹³C NMR (100 MHz, DMSO) δ 156.4 (C), 137.0 (C), 134.7 (C), 128.9 (CH), 128.7 (CH), 127.4 (CH), 126.7 (CH), 126.3 (CH), 120.1 (C), 105.1 (CH), 23.7 (CH), 20.6 (CH₃).

HRMS-APCI (m/z) [M+H]⁺ calcd for C₁₇H₁₉O₂, 255.1380; found, 255.1376.

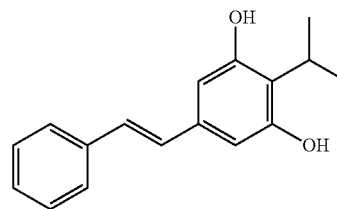
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The invention claimed is:

1. A pharmaceutical composition comprising:

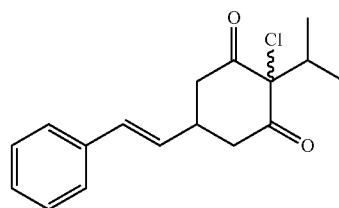
a compound consisting of

at least 99% by weight of a compound of Formula (I)



wherein the compound is at least 95% form 1, characterized by an X-ray powder diffraction pattern having peaks at 15.0, 17.8, 19.1, 20.2, 21.5, 22.4, 23.3, 24.5, 26.2 and 27.9 degrees 2θ (±0.1° 2θ); and

less than or equal to 1% impurities; wherein the impurities include at least a compound of the Formula



and

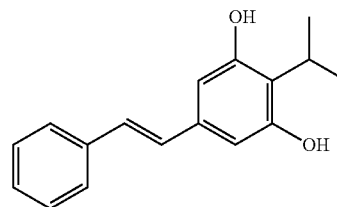
a pharmaceutically acceptable excipient.

2. A process for preparing the compound of the composition of claim 1, comprising crystallizing a compound of Formula (I) using methanol and water.

3. The process of claim 2, wherein the process comprises seeding.

4. The compound of the composition of claim 1 prepared by a process comprising crystallizing a compound of Formula (I) using methanol and water.

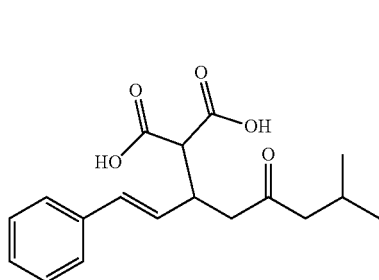
5. A compound of Formula (I)



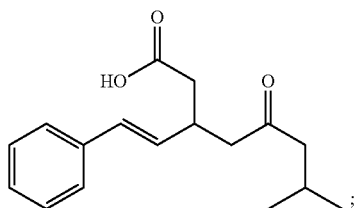
or a salt or solvate thereof; wherein the compound or a salt or solvate thereof, is at least 99% pure by weight and wherein the compound is at least 95% form 1 characterized by an X-ray powder diffraction pattern having peaks at 15.0, 17.8, 19.1, 20.2, 21.5, 22.4, 23.3, 24.5, 26.2 and 27.9 degrees 2θ (±0.1° 2θ); prepared by a process comprising:

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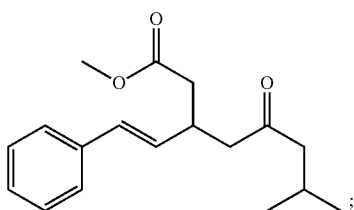
a) decarboxylating a compound of Formula (VI)



in the presence of a base selected from trimethylamine, imidazole, pyridine, and lutidine (2,6-dimethylpyridine) to form a compound of Formula (V) or a salt thereof

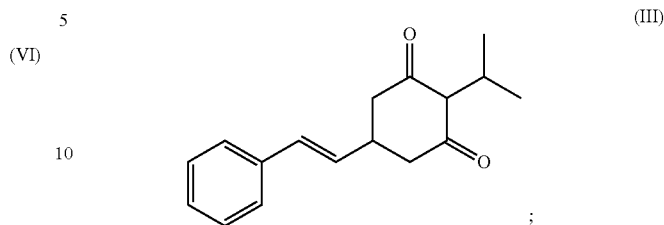


b) esterifying the compound of Formula (V) or a salt thereof with methanol and aqueous hydrochloric acid to form a compound of Formula (IVa)

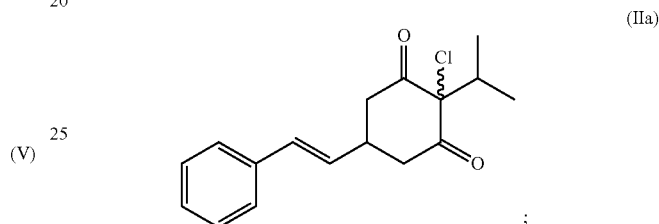


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c) cyclizing the compound of Formula (IVa) with potassium tert-butoxide to form a compound of Formula (III)



d) halogenating the compound of Formula (III) with a halogenating agent selected from 1,3-dichloro-5,5-dimethylhydantoin, N-chlorosuccinimide, and trichloroisocyanuric acid, to form a compound of Formula (IIa)



and

e) aromatizing the compound of Formula (IIa) in the presence of an additive reagent selected from tetrabutylammonium bromide, benzyltriethylammonium chloride, tetrabutylammonium chloride, tetraethylammonium chloride, to form the compound of Formula (I) or a salt or solvate thereof.

6. The compound of claim 5, wherein the compound is about 96% form 1.

7. The compound of claim 5, wherein the compound is about 97% form 1.

8. The compound of claim 5, wherein the compound is about 98% form 1.

9. The compound of claim 5, wherein the compound is about 99% form 1.

10. The compound of claim 5, wherein the compound is at least 99% form 1.

11. The compound of claim 5, wherein the process further comprises:

f) crystallizing the compound of Formula (I) using methanol and water.

12. The compound of claim 11, wherein the crystallizing comprises seeding.

* * * * *